

A Review of Body Biasing as a Low Power Design Technique

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Abstract — Body biasing is a technique used in integrated circuit (IC) design to modulate the threshold voltage (V_{th}) of a MOS transistor by altering its body (or bulk) voltage. As the demand for faster and lower-power electronics continues to grow, researchers have explored various methods to optimize circuit performance and energy efficiency without sacrificing area. Among these, body biasing has played a critical role in enabling fine-grained control of both power dissipation and switching speed. This survey explores three primary body biasing approaches i.e Forward Body Biasing (FBB), Reverse Body Biasing (RBB), and Adaptive Body Biasing (ABB). FBB decreases V_{th} thereby increasing turn-on speed at the cost of higher leakage power. RBB, conversely, increases V_{th} effectively reducing leakage but slowing down circuit operation. ABB dynamically switches between FBB and RBB depending on performance or power demands, aiming to strike a balance between speed and energy efficiency. Body biasing has often been implemented in conjunction with other low-power design techniques such as Dynamic Voltage and Frequency Scaling (DVFS), Power Gating, and Multi-Threshold CMOS (MTCMOS) to maximize their benefits. Despite being overshadowed by these techniques as transistors get smaller, body biasing still remains relevant especially in specific applications, technology platforms like FD-SOI, and scenarios requiring fine-tuned control. This survey reviews its evolution, technical foundations, advantages, limitations, and its evolving role in the landscape of modern low-power IC design.

I. INTRODUCTION

MOSFETs are the fundamental building blocks of modern digital circuits. While CMOS transistors are usually thought of as having three terminals in common practice namely the source, gate and drain, the body also often plays a crucial role in influencing the behavior of a transistor. In fact, it is sometimes referred to as the “fourth gate.” Under normal conditions, it’s expected that the voltage difference between the source and the drain is equal to zero, however this is often not the case. This phenomenon is referred to as the body effect and it plays a significant role in modulating the threshold voltage [8]. Early CMOS designs typically grounded the body, but engineers later discovered that

dynamically adjusting the body voltage could take advantage of this effect giving rise to body biasing.

The three main biasing techniques are Forward Biasing (FBB), Reverse Biasing (RBB) and Adaptive Biasing (ABB) [8]. In FBB, the potential difference between the source and the body is low, lowering the voltage needed to form the inversion layer. RBB is the opposite where higher voltage is needed to form the inversion layer because of the higher source-to-body voltage $|V_{SB}|$. ABB switches between FBB and RBB based on the desired performance. These techniques are implemented differently in NMOS and PMOS devices, each with distinct optimal operating points depending on the desired gain and acceptable trade-offs. This paper is a survey of these techniques from how they work, their impact and relevance as technology continues to advance.

II. BACKGROUND: THE BODY EFFECT AND BIASING TECHNIQUES

In MOSFETs, As shown by the figure below, a parasitic diode is formed between the doped regions (drain/source) and the bulk. These diodes influence the operation of the device. For the devices to operate correctly, these diodes should always be reverse biased.

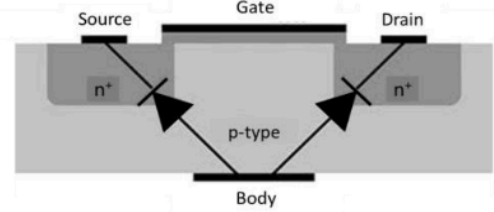


Fig. 1. A representation of parasitic diodes in a nMOS transistor. Adapted from [8]

Normally, the body is expected to be connected to the source but this is not useful because there are thousands or millions of transistors in a MOSFET device all connected to the same substrate/bulk. Connecting the body to the source in this case results in connecting all the terminals to each other resulting in a useless circuit. Since the body and the source

are connected to different terminals, $V_{SB} \neq 0$ hence the body effect. Taking this into consideration, the correct expression of the threshold voltage becomes;

$$V_{TH} = V_{TH0} + \gamma(\sqrt{2\Phi_F + V_{SB}} - \sqrt{2\Phi_F}) \quad (1)$$

Where V_{TH0} is the device threshold voltage with no body bias applied, $2\Phi_F$ is the surface potential at strong inversion, and γ is the body effect coefficient [11].

Threshold voltage affects power dissipation as leakage current and delay depends on V_{TH} as show [14]:

$$I_{sub} = I_0 \cdot e^{\frac{V_{GS} - V_{TH0} - \lambda \cdot V_{SB} + \eta \cdot V_{DS}}{n \cdot V_T}} \cdot \left(1 - e^{-\frac{V_{DS}}{V_T}}\right) \quad (2)$$

$$t_p \propto \frac{C_L \cdot V_{DD}}{I_{DS}} \approx \frac{C_L \cdot V_{DD}}{A \cdot (V_{DD} - V_{TH})} \quad (3)$$

where I_0 is the drain-source current (I_{DS}) when $V_{GS} = V_{TH0}$, V_T is the thermal voltage ($k \cdot T/q$), η is drain-induced barrier lowering (DIBL) coefficient, n is the sub-threshold slope, C_L is the fan-out load capacitance, and A is a constant factor.

Even Though the body effect is no longer nontrivial in sub-100nm technologies, strong body factor technologies like Bulk CMOS and Fully-Depleted-Silicon-On-Insulator (FD-SOI) circuits still benefit from manipulating their body voltages.

A. Forward Body Biasing (FBB)

A device operates in forward bias if the body voltage is more positive in relation to the source voltage (or more negative for PMOS). This will decrease the depletion width hence lowering the voltage needed to form the inversion layer. This means that the minimum voltage required to turn on the transistor i.e the threshold voltage reduces. In a NMOS device, this effect is achieved when V_B is greater than V_S and when V_B is less than V_S in a PMOS device. FBB gives the transistors a high operational frequency while also improving the short channel effect [1] [2]. The shorter depletion width however means more electrons can travel through the P-N junction even when the device is not operating, increasing the subthreshold current leakage I_{OFF} as shown by the results from a study by N. Kamal et al.

	ZBB	FBB	RBB	DTMOS
Leakage current	0.7 nA	59.0 nA	0.07 nA	0.7 nA
Power consumption n	0.8 nW	1100 nW	0.08 nW	0.8 nW

Fig. 2. A comparison of leakage current and power dissipation in a zero, forward and reverse biased device. Adapted from [8]

FBB also makes the devices more sensitive to any changes in the subthreshold voltage improving scalability as technology advances [10]. This however also means that the devices became more sensitive to noise [4]. FBB can be applied in several ways depending on performance and leakage requirements. Swapped FBB forward-biases the body-source junction continuously, keeping the transistor always on. Due to high leakage, it's limited to only ultra-low-voltage circuits where switching speed outweighs leakage concerns.. Clocked FBB applies forward bias only during active clock phases and is used to balance speed and leakage. Hybrid FBB combines both approaches, applying bias during active clock operations and reducing it in standby to tune performance adaptively. Dynamic Body Biasing (DBB) sometimes called Dynamic Threshold MOS (DTMOS) involves short circuiting the body to the gate that way the device is forward biased when on and zero biased when off to minimize leakage [4].

For some technologies like dual-well and merged triple-well configurations, FBB is suspected to be one of the phenomena that can trigger latch-up events due to the possible activation of the positive feedback loop formed by a parasitic PNP structure because of the voltage drop associated with FBB [4] [11]. Because of these limitations, forward biasing is typically limited to ~ 0.3 V to ~ 0.5 V.

B. Reverse Body Biasing (RBB)

In reverse bias, the depletion width increases as the potential difference between the body and the source increases. This in turns leads to increase in voltage needed to form the inversion layer hence the high V_{TH} . In PMOS this is achieved when V_S is less than V_B and when V_S is greater than V_B in NMOS as shown by the configuration below.

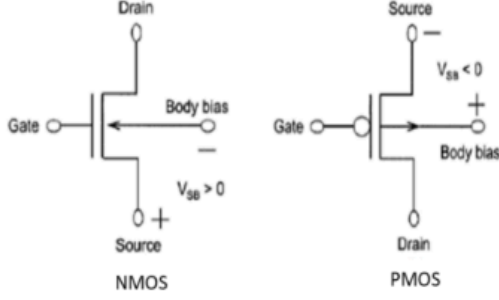


Fig. 3. A representation of how transistors are reverse biased.
Adapted from [8]

By increasing the threshold voltage, RBB reduces subthreshold leakage current, making it effective for leakage power reduction, especially in idle or standby modes. This technique improves energy efficiency and is commonly used in sleep modes or always-on circuitry where power savings outweigh the need for speed [3] [9]-[11]. RBB however creates stronger electric fields at the PN junction leading to a significant increase in band-to-band tunneling increasing I_{OFF} in heavily doped substrates making it less effective in minimizing leakage power dissipation [10] [13]. It also increases the delay time lowering operational frequency. The longer turn-on times associated with RBB may also be a contributing factor in increasing I_{ON} [3].

Unlike FBB, reverse biasing limits the devices' sensitivity to changes in subthreshold voltage, V_{SUB} making it less useful as the ranges of V_{SUB} become more limited as technology advances [10].

C. Adaptive Body Biasing (ABB)

This technique aims to combine the best of FBB and RBB to optimize performance and power consumption by dynamically adjusting the body voltage of transistors in real time based on performance demands, temperature, or process variation. With ABB, when high performance is needed, the device is forward biased and when it's idle or no need for high performance, the device is switched to reverse bias mode. It is the most widely used technique of the three due its effectiveness and scalability.

ABB not only improves energy efficiency but also compensates for variations in manufacturing processes that affect transistor speed and leakage differently across dies and regions of a chip. For instance, slower dies or underperforming blocks can be selectively forward-biased to meet timing requirements, while faster dies can be reverse-biased to minimize power without sacrificing performance thus increasing the number of acceptable dies [1].

A study by A. Srivastava and C. Zhang found that ABB was able to reduce up to 90% of current leakage in low threshold MOSFET circuits making it particularly effective for subthreshold or near-threshold designs where leakage is a dominant concern.

On the other hand, ABB requires additional on-chip circuitries like bias generators, control logic and sensors, which introduce area, cost and design complexities. These added components also introduce extra verification and reliability challenges, particularly in deeply scaled nodes where body effect control is already limited [12].

III. BODY BIASING WITH OTHER LOW DESIGN TECHNIQUES

As technology scales down, the body effect becomes less and less of a concern making body biasing less useful. Body biasing however still stands at a good place when it comes to its integration with other low power techniques to make them more effective. Several research have proposed some level of body biasing to be integrated with the designs below to make these techniques work better;

1. *Body Biasing and Dynamic Voltage and Frequency Scaling (DVFS)*: DVFS reduces dynamic and static power by lowering supply voltage and clock frequency during low-performance requirements. One study found that integrating ABB with DVS increases its average energy reduction from 23% to 48% [6] while another observed an increase of 18% from 11% when integrating ABB and DVFS [15]. By dynamically adjusting both supply voltage (DVFS) and body bias (ABB), designers can achieve lower power consumption than with DVFS alone, especially by reducing leakage in low-voltage modes and boosting performance when needed.
2. *Body Biasing and Power Gating*: Power gating disconnects power from idle circuit blocks to reduce leakage power. While ABB plays an important role in reducing both dynamic and static power consumption, it has also shown significant energy efficiency results during wake up times in a power gated system [16].
3. *Body Biasing and Multi-Threshold CMOS (MTCMOS)*: MTCMOS uses transistors with different threshold voltages depending on the desired speed or leakage control. High V_{th} is used for high speed and low V_{th} for low leakage. Combining ABB with MTCMOS allows for both coarse (via device selection) and fine (via bias adjustment) control of leakage and performance [17].

IV. TECHNOLOGY SCALING VS. BODY BIASING

For technologies below 100 nm, especially in bulk CMOS, the effect of body biasing decreases tremendously. This is due to the reduced body factor, which limits how much the body terminal can influence the channel potential. At the 90 nm node, body biasing gain is already reduced by a factor of approximately 4 $\times$, and by 65 nm, only a 2 $\times$ modulation effect is observed. Below 40 nm, this gain becomes minimal, and in sub-20 nm technologies, it is effectively negligible [12]. As a result, the cost and complexity of implementing body biasing often outweigh its diminishing benefits at advanced nodes.

At the same time, short-channel effects (SCEs) such as threshold voltage roll-off and drain-induced barrier lowering (DIBL) have become increasingly nontrivial with shrinking channel lengths. These effects reduce electrostatic gate control and intensify leakage. While FBB was once leveraged to counteract SCEs by enhancing drive strength and improving subthreshold characteristics, its effectiveness has been compromised by reduced body influence and the risk of increased leakage currents in scaled technologies as shown by this study [1] [19].

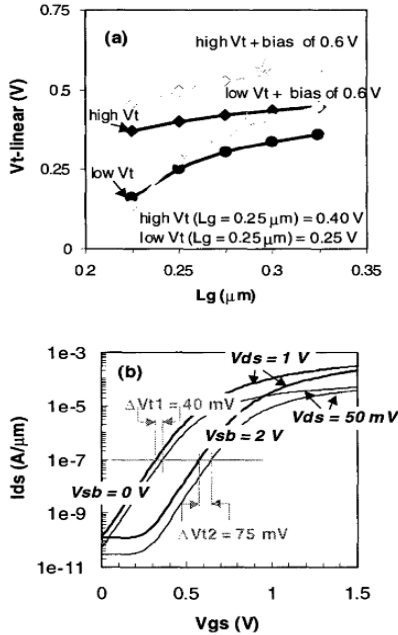


Fig. 4. Results on how biasing effects V_{th} roll-off (Top) and DIBL (Bottom). Adapted from [19]

In this study done on low threshold transistors, the results show that V_{th} roll-off worsens as body bias is applied and there is an increase in DIBL as the potential difference between the source and the body terminals increases.

In emerging non-planar technologies like FinFETs and Gate-All-Around FETs, body access is either eliminated or poorly coupled to the channel, further limiting biasing gain. These devices rely instead on work function engineering and channel doping for threshold tuning [18]. Additionally, bias limits coupled with being overshadowed by more scalable power management techniques like power gating and DVFS has made body biasing less practically beneficial.

V. BODY BIASING TODAY

Despite its diminishing use, body biasing still retains practical use in select technologies where body control is still stronger and can be leveraged at finer granularity.

a) FD-SOI-Based Systems

FD-SOI is the leading technology for body biasing, offering a wide range of bias voltages and dynamic control not possible in bulk CMOS or FinFET. Its well-isolated body and ultra thin buried insulator layer gives it a greater threshold voltage sensitivity enabling strong biasing gain and fine-grained adaptive control [3] [5] [12]. Leading semiconductor manufacturers like Samsung, GlobalFoundries and STMicroelectronics have started considering the integration of FD-SOI technology on a commercial level in their AI and Edge Computing, automotive, communications, IoT and consumer electronics.

b) Bulk Cryo-CMOS

For bulk CMOS, especially those that operate at cryogenic temperatures, body biasing, especially FBB can still be leveraged to curb the significant increase in threshold voltages as the temperature increases [7]. Overwater et al. demonstrate that at cryogenic temperatures (e.g., 4.2 K), the increased diode turn-on voltage mitigates leakage risks typically associated with FBB at room temperature, enabling its safe application even with full bias voltages. This permits the use of simpler bulk CMOS circuits in quantum computing and space exploration systems, which benefit from enhanced speed (1.6 $\times$ increase), reduced energy-delay product (2.3 $\times$ improvement), and restored circuit functionality under extreme conditions. Importantly, these gains are achieved without the need for beyond-supply voltages or costly FD-SOI processes, thus reaffirming body biasing's viability in cryo-CMOS bulk designs today.

VI. RECENT ADVANCES AND RESEARCH DIRECTIONS

Thanks to the return of FD-SOI technology and advancement in 3D technology, researchers have started to revisit body biasing particularly, Adaptive Body Biasing. Within the last 5 years;

- I. GlobalFoundries has developed new all-in-one adaptive body-bias subsystems that enable both forward and reverse body biasing for consumer and ultra-low-power IoT applications. These IP blocks include on-chip body-bias voltage generators and closed-loop regulation to dynamically compensate for process, voltage, and temperature (PVT) variations, maximizing FD-SOI's body-biasing capabilities.
- II. Efforts exploring ABB in cryogenic environments have emerged demonstrating the effectiveness of adaptive back-gate voltage control for maintaining device performance and energy efficiency at ultra-low temperatures [7] [20].
- III. There's growing interest in applying ABB in heterogeneous 3D-stacked systems, where body bias could be applied per-die or per-layer, offering local thermal and performance adaptation. This is an active research area, especially for logic-memory stacks and chiplets.
- IV. Though not popular, there are some academic and industrial efforts being put into new characterization methodologies to model body bias effects across corner cases and voltage ranges, allowing ABB-aware optimization during RTL-to-GDS flows [17].

VII. THE FUTURE OF BODY BIASING

In recent years, the quest for faster, low-power and cheaper electronics has seen a big shift from trying to make traditional semiconductor materials better via dimensional scaling to exploring new materials altogether. While silicon is yet to be replaced by these new emerging technologies, the relevance of body control continues to diminish.

In 2D nanoelectronics for example, materials like graphene, MoS₂, and other transition metal dichalcogenides are atomically thin and often lack a true "bulk" substrate beneath the channel making body effect negligible. Field Effect transistors built out of these materials do not have a body terminal since they have great gate control over the channel.

Other materials like wide-bandgap semiconductors, operate at higher threshold voltages and reduced sensitivity to body bias compared to silicon. Besides, WBG based devices like GaN HEMTs are often normally-off (enhancement-mode) eliminating the need for body control. Their device architectures also do not always support conventional body control as the channel is often formed at a heterojunction or interface, not modulated by a traditional substrate bias [21]. Narrow-bandgap materials on the other hand have stronger electrostatic sensitivity, which could enhance the body effect.

However, this also means more pronounced leakage currents and DIBL effects, making body biasing less useful or even deteriorative.

In other technologies like superconductors and vacuum channel devices, the concept of body effect is non-existent and by extension, body biasing. This highlights a critical limitation in the scalability of body biasing as we move into more exotic and non-traditional semiconductor technologies, the relevance of body biasing may diminish.

Nonetheless, crucial technologies like FD-SOI, cryo-CMOS, near-threshold and sub-threshold computing will always send designers back to body biasing especially ABB, suggesting that while it may not extend to all future devices, it will remain a powerful tool in silicon-based and hybrid systems for the foreseeable future. The still widespread use of technologies that rely on above 100 nm bulk CMOS also means that body biasing is yet to be a concept of the past.

Finally, though these advances have reignited interest in body biasing, the long-term trajectory of the field remains uncertain. As technology nodes continue to scale below 10 nm and architectures move further away from bulk CMOS towards devices like FinFETs, gate-all-around FETs and non-silicon platforms, the structural foundations that enable strong body control are diminishing. Given this shift, I believe body biasing has entered a phase of selective relevance rather than widespread adoption. The technique will likely remain dormant across mainstream logic technologies, reappearing occasionally in niche or specialized domains where its advantages are uniquely beneficial. Unless future technologies are explicitly designed to support body control, the field of body biasing is unlikely to evolve linearly with CMOS but rather survive through targeted applications and design-specific integrations.

VIII. CONCLUSION

Body biasing, especially Adaptive Body Biasing (ABB) still continues to offer designers a flexible means of meeting energy and performance targets despite the diminishing body effect due to scaling in CMOS. In environments where power sensitivity is critical, integrating body biasing with the more scalable techniques have proved to be more beneficial than if these techniques were applied independently.

Other than ABB, studies have also shown that Forward Body Biasing (FBB) was a very widely used and effective biasing technique due to its ability to make CMOS more scalable and sensitive to changes in the subthreshold voltage. Reverse Body Biasing (RBB) was the least used biasing technique independently though it still proved very useful in

technologies where the need for power management outweighed speed.

Despite the diminishing use and the rise of more dominant techniques in modern ultra-scaled technologies, body biasing remains relevant, not as a universal solution, but as a complementary tool for specialized and performance-aware design.

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